Scenario -> Shopping Cart

❌ Weak: "Test the cart functionality."

O/p using above Prompt ->

Please share the website or application URL where you want me to test the cart functionality.

I'll help verify these key scenarios:

Test Case	Expected Result
Add item to cart	Item appears in the cart
Add multiple items	All selected items appear correctly
Update quantity	Quantity and total price update correctly
Remove item	Item is removed and total recalculates
Empty cart	Empty-cart message is displayed
Refresh the page	Cart contents behave as expected
Verify total	Subtotal, discounts, tax, and shipping are calculated correctly
Proceed to checkout	User can continue to checkout
Handle invalid quantity	Zero, negative, or excessive quantities are rejected appropriately

Send me the URL or the relevant application code, and I'll help test the actual behavior and identify any issues.

I can also create an image illustrating the cart-testing workflow.

Using ICEPOT method: ->

I:-> Write Test Cases to test the cart functionality

C: -> Cart has multiple components like Product line items, Product Prices, Delete Product from Cart button, Checkout button, Out of Stock Products, Product quantities

E: -> Consider example like -> Verify Product quantities can be changed, products can be deleted, Navigation to checkout page, count line items etc.

P: -> Act as a Test Engineer

O: -> Provide output in the form of a table with proper headers like, Test Case ID, Pre-condition, Test Case name, Description, Steps, Expected Result, etc.

T: -> Test Cases should have the maximum coverage

O/P using above Prompt ->

Below is a comprehensive cart functionality test suite prepared from the perspective of a **Test Engineer**. It covers positive scenarios, negative scenarios, boundary conditions, UI behavior, price calculations, inventory validation, checkout navigation, session management, accessibility, and security.

Test Cases: Shopping Cart Functionality

1. Objective

To verify that the shopping cart works correctly when users add products, view product details, modify quantities, remove products, check prices, handle out-of-stock products, and proceed to checkout.

The test cases also cover data validation, error handling, cart persistence, usability, performance, and security.

2. Test Environment and Assumptions

Application under test: E-commerce website or application.

Prerequisites:

- The application is accessible.
- Products are available in the product catalog.
- Users can add products to the cart.
- The cart page and checkout page are available.
- Test accounts are available for guest and authenticated-user scenarios.

- Product prices, inventory quantities, and cart rules are configurable or known.

Sample test data:

Product	Product ID	Unit Price	Available Stock	Status
Product A	PROD001	$10.00	10	In Stock
Product B	PROD002	$25.00	5	In Stock
Product C	PROD003	$15.00	0	Out of Stock
Product D	PROD004	$50.00	2	Limited Stock
Product E	PROD005	$20.00	100	In Stock

For the test cases below, assume that the cart identifies a line item using the product and its selected variant. Products with different variants are treated as distinct line items. Adjust this assumption if the application's requirements specify different behavior.

Priority definitions:

Priority	Meaning
P0	Critical: Core cart or checkout functionality is blocked.
P1	High: Important functionality is incorrect or unreliable.
P2	Medium: A secondary, usability, or edge-case issue exists.

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3. Cart Display and Product Line Items

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-001	Product A is available and the cart is empty.	Verify cart displays a single product	Validate the cart when one product has been added.	1. Open the product page for Product A. 2. Add Product A to the cart.	Product A appears as a line item with the correct name, price, quantity,	P0

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
				3. Open the cart.	and other required details.	
TC-CART-002	Products A and B are available.	Verify cart displays multiple products	Validate the display of multiple distinct products.	1. Add Product A to the cart. 2. Add Product B to the cart. 3. Open the cart.	Both products appear correctly, without missing or unintended duplicate line items.	P0
TC-CART-003	A product has been added to the cart.	Verify product details in cart	Check the information displayed for each line item.	1. Add Product A to the cart. 2. Open the cart. 3. Inspect the product name, image, selected variant, SKU, and other applicable attributes.	Displayed product details match the selected catalog product and variant. Required information is visible and accurate.	P1
TC-CART-004	Two distinct products are in the cart.	Verify number of line items	Validate the displayed line-item count.	1. Add Product A. 2. Add Product B. 3. Open the cart and inspect the line-item count.	The cart contains two distinct line items, and the displayed line-item count is correct if the application provides one.	P1
TC-CART-005	The cart contains Product A	Verify total item	Check whether the cart badge	1. Add Product A twice or set its quantity to	If the badge represents total units, it displays	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
	with quantity 2 and Product B with quantity 3.	quantity indicator	reflects the required quantity-counting rule.	2. 2. Add Product B and set its quantity to 3. 3. Inspect the cart badge.	5. If it represents distinct line items, it displays 2. The behavior matches the documented requirement.	
TC-CART-006	Product A is available.	Verify adding the same product multiple times	Determine whether repeated additions update an existing line item or create duplicates.	1. Add Product A to the cart. 2. Navigate back to Product A. 3. Add Product A again. 4. Open the cart.	The cart follows the defined duplicate-product rule. For a product with identical variants, the expected default is one line item with the quantity increased to 2.	P1
TC-CART-007	Product A is available in Red and Blue variants.	Verify products with different variants appear separately	Ensure that selecting different product variants creates the appropriate line items.	1. Add Product A in Red. 2. Add Product A in Blue. 3. Open the cart.	The variants appear as separate line items, with the correct variant attributes, prices, and quantities.	P1
TC-CART-008	The cart is empty.	Verify empty-cart message	Confirm the correct state is displayed when no	1. Open the cart without adding any products.	An appropriate empty-cart message or design is displayed. No stale product	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
			products are present.		details are visible, and available navigation or shopping actions work correctly.	

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4. Product Quantity Validation

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-009	Product A is in the cart with quantity 1.	Verify increasing product quantity	Check whether the user can increase a product's quantity.	1. Open the cart. 2. Click the plus button once.	Quantity changes from 1 to 2. The displayed line subtotal and cart subtotal update correctly.	P0
TC-CART-010	Product A is in the cart with quantity 2.	Verify decreasing product quantity	Check whether the user can decrease a product's quantity.	1. Open the cart. 2. Click the minus button once.	Quantity changes from 2 to 1, and all affected price totals are recalculated correctly.	P0
TC-CART-011	Product A is in the cart and quantity editing is supported.	Verify direct quantity entry	Validate quantity changes using the quantity input field.	1. Open the cart. 2. Select the quantity field. 3. Replace its value with 4. 4. Trigger the	The quantity is updated to 4, and all affected totals reflect the new quantity.	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
				supported update action.		
TC-CART-012	Product A is in the cart with quantity 2. The minimum permitted quantity is 1.	Verify minimum quantity boundary	Ensure the quantity cannot fall below the allowed minimum.	1. Reduce the quantity to 1. 2. Attempt to decrease it again.	The application prevents an invalid quantity. If zero quantity is the designated removal action, the product can be removed only through the supported behavior.	P1
TC-CART-013	A product is in the cart and direct quantity editing is supported.	Verify zero and negative quantities	Validate handling of invalid nonpositive quantities.	1. Enter 0. 2. Trigger the update. 3. Repeat with -1.	Invalid quantities are rejected or normalized according to the defined rules. Negative quantities are never accepted, and no incorrect totals are produced.	P1
TC-CART-014	A product is in the cart and quantity editing is supported.	Verify nonnumeric and decimal quantity values	Validate quantity-field input restrictions.	1. Enter alphabetic characters. 2. Attempt to enter special characters. 3. Enter a decimal value	Only valid quantity values are accepted. For products sold in whole units, nonnumeric and fractional values are rejected or	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
				such as 1.5. 4. Trigger the update.	normalized according to the field's validation rules.	
TC-CART-015	A product has a configured maximum quantity limit.	Verify maximum permitted quantity	Check the upper boundary of the quantity field.	1. Add the product to the cart. 2. Attempt to set its quantity to the maximum permitted value. 3. Attempt to exceed that value.	The permitted maximum is accepted. Values above the configured limit are prevented or rejected with an appropriate message.	P1
TC-CART-016	Product A has 10 units available.	Verify quantity cannot exceed available stock	Check stock-limit validation during quantity updates.	1. Add Product A to the cart. 2. Attempt to increase its quantity to 11.	The application prevents the user from ordering more than the available stock, or clearly identifies the stock restriction and blocks checkout until resolved.	P0
TC-CART-017	Product A costs $10.00 and is in the cart.	Verify line subtotal after quantity change	Validate the calculation for one product line.	1. Set Product A's quantity to 3. 2. Inspect its line subtotal.	The line subtotal is $30.00 before any applicable line-level discounts, taxes,	P0

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
					or other adjustments.	
TC-CART-018	Product A costs $10.00 and Product B costs $25.00. Both are in the cart.	Verify cart subtotal after quantity change	Ensure changes to one quantity update the overall subtotal.	1. Set Product A quantity to 2. 2. Set Product B quantity to 1. 3. Inspect the subtotal.	The subtotal is $45.00 before applicable discounts, taxes, shipping, and other adjustments.	P0
TC-CART-019	Product A is in the cart and has sufficient stock.	Verify rapid quantity updates	Test multiple quick interactions with quantity controls.	1. Click the plus button several times rapidly. 2. Wait for all updates to complete. 3. Inspect the displayed quantity and totals.	All accepted updates are processed correctly. The displayed quantity, cart state, and totals remain consistent, without lost updates or duplicate requests that produce incorrect results.	P1

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5. Product Prices and Cart Calculations

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-020	Product A is displayed in the catalog with a price of $10.00.	Verify product unit price	Validate the price displayed in the cart.	1. Add Product A to the cart. 2. Open the cart. 3. Compare the unit price with the applicable catalog price.	The cart displays the correct unit price for the selected product and variant, subject to the application's pricing rules.	P0
TC-CART-021	Product A has a unit price of $10.00 and quantity 3.	Verify line subtotal calculation	Ensure line subtotal uses the correct unit price and quantity.	1. Add Product A. 2. Set its quantity to 3. 3. Inspect the line subtotal.	The line subtotal is $30.00 before applicable line-level adjustments.	P0
TC-CART-022	Product A costs $10.00 and Product B costs $25.00.	Verify total cart subtotal	Validate that the subtotal equals the sum of the cart's line subtotals.	1. Add Product A with quantity 2. 2. Add Product B with quantity 1. 3. Inspect the subtotal.	The subtotal equals $45.00 before discounts, taxes, shipping, and other adjustments.	P0
TC-CART-023	A product has a configured sale price and original price.	Verify discounted product price	Check the display and calculation of sale prices.	1. Add the discounted product. 2. Open the cart. 3. Inspect the original price, sale price, discount	The displayed price follows the pricing rules. Any advertised savings and line subtotal are calculated accurately, with no unintended	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
				information, and subtotal.	use of the original price in place of the applicable sale price.	
TC-CART-024	The application supports promotional codes. A valid code is available.	Verify promotional discount calculation	Check how an eligible discount affects the cart.	1. Add eligible products. 2. Apply a valid promotional code. 3. Inspect the discount and resulting totals.	The valid discount is applied once according to its rules. The displayed discount and adjusted total are correct.	P1
TC-CART-025	The application calculates tax and shipping.	Verify tax and shipping calculations	Ensure applicable charges are calculated and presented correctly.	1. Add products to the cart. 2. Enter any required location information. 3. Inspect available tax, shipping, and total calculations.	Charges are calculated and displayed according to the application's pricing rules. Estimates are distinguished from finalized checkout amounts where required.	P1
TC-CART-026	A product has a price that produces fractional	Verify currency rounding	Check monetary precision and rounding behavior.	1. Add the product. 2. Change its quantity several times. 3. Compare	Currency amounts use the supported precision and rounding rules consistently. No	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
	currency calculations.			the displayed totals with the defined calculation rules.	unexpected floating-point artifacts appear in displayed prices.	
TC-CART-027	The application supports a configured currency.	Verify currency formatting	Check the currency symbol, decimal precision, and formatting.	1. Add a product. 2. Open the cart. 3. Inspect all displayed monetary values.	All applicable monetary values use the correct currency and formatting. No inconsistent or unintended currency symbols are shown.	P2
TC-CART-028	Product A is in the cart. Its catalog price can change.	Verify product price changes after adding to cart	Validate how the cart handles a changed catalog price.	1. Add Product A to the cart. 2. Change its catalog price in the test environment. 3. Refresh or reopen the cart.	The cart follows the documented price-locking or price-refresh policy. If the price changes, the updated amount is clearly shown and all totals remain consistent.	P1

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6. Delete Product from Cart

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-029	Products A and B are in the cart.	Verify deleting one product	Ensure a specific product can be removed.	1. Open the cart. 2. Click the remove button for Product A. 3. Inspect the remaining cart contents.	Product A is removed. Product B remains in the cart with its correct details and quantity.	P0
TC-CART-030	The cart contains only Product A.	Verify deleting the last product	Check cart behavior after removing its final line item.	1. Open the cart. 2. Remove Product A.	The cart displays its empty state. No stale line items or incorrect totals remain.	P0
TC-CART-031	Products A and B are in the cart.	Verify line-item count after deletion	Ensure removing a product updates the line-item count.	1. Verify the cart contains two distinct line items. 2. Remove Product A. 3. Check the line-item count.	The number of distinct line items decreases from 2 to 1. Any cart badge follows its defined counting convention.	P1
TC-CART-032	Product A has quantity 2 and Product B has quantity 1.	Verify totals after product deletion	Check recalculation after removing a line item.	1. Add both products. 2. Remove Product A. 3. Inspect the remaining subtotal and totals.	Product A's amount is excluded from the subtotal. All applicable discounts, taxes, shipping, and totals are recalculated	P0

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
					according to the application's rules.	
TC-CART-033	The application displays a removal confirmation dialog. Product A is in the cart.	Verify cancelling a product removal	Ensure cancellation does not delete the product.	1. Click Remove for Product A. 2. Select Cancel in the confirmation dialog.	Product A remains in the cart, its quantity is unchanged, and all totals remain correct.	P2
TC-CART-034	A product is in the cart. Network failure can be simulated.	Verify behavior when removing a product fails	Validate error handling when the server cannot complete a removal.	1. Simulate a network or server failure. 2. Attempt to remove the product. 3. Inspect the cart state.	The application presents appropriate feedback. It does not falsely indicate that removal succeeded if the server rejected the change. The user can recover or retry without creating inconsistent cart data.	P1

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7. Out-of-Stock Products and Inventory

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-035	Product C has zero stock and backorders are not allowed.	Verify adding an out-of-stock product	Ensure unavailable products cannot be purchased.	1. Open Product C's page. 2. Attempt to add it to the cart.	The application prevents adding the product for purchase, or follows its defined unavailable-item workflow. It does not silently create a purchasable line item for an unavailable product.	P0
TC-CART-036	Product A is in the cart and initially has stock.	Verify an item becomes out of stock while in the cart	Validate inventory changes after a product has already been added.	1. Add Product A to the cart. 2. Change its available stock to zero through the test environment. 3. Refresh or reopen the cart.	The application detects or communicates the inventory change. The user cannot proceed as though the item were available when backorders are not permitted.	P0
TC-CART-037	Product A has 10 available units.	Verify adding more units than available stock	Test a quantity that exceeds inventory.	1. Add Product A to the cart. 2. Set its quantity to 11. 3. Attempt	The quantity is rejected or constrained to the supported maximum. The user receives appropriate	P0

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
				to update the cart.	feedback, and invalid quantities are not accepted by the server.	
TC-CART-038	Product D has only 2 units available.	Verify ordering the remaining available stock	Check quantity handling near the stock limit.	1. Add Product D to the cart. 2. Set its quantity to 2. 3. Attempt to increase it to 3.	Quantity 2 is accepted. The application prevents quantity 3 or clearly identifies the stock restriction. Totals remain correct.	P1
TC-CART-039	At least one product is out of stock.	Verify out-of-stock status in cart	Ensure product availability is visible and understandable.	1. Add a product while it is available. 2. Make the product unavailable through the test environment. 3. Open the cart.	The cart presents the required availability status or an actionable message. The user is not misled into believing that an unavailable product can definitely be purchased.	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-040	A product is in the cart; inventory can change before checkout.	Verify stock validation before checkout	Confirm the application revalidates inventory before proceeding with a purchase.	1. Add an available product to the cart. 2. Reduce its available stock to zero. 3. Attempt to proceed to checkout.	The application revalidates stock and blocks or appropriately adjusts the unavailable item according to business rules. Clear feedback is presented, and no invalid purchase is accepted.	P0
TC-CART-041	A product supports backorders or preorders.	Verify purchasing a backorder-eligible product	Check the behavior for products that are unavailable for immediate shipment but remain orderable.	1. Add the eligible product to the cart. 2. Open the cart. 3. Proceed to the next supported purchase step.	The product follows the documented backorder or preorder policy. Any applicable availability, fulfillment, or estimated-delivery information is presented accurately.	P2

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8. Checkout Button and Navigation

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-042	The cart contains at least one valid, purchasable product.	Verify checkout button availability	Check whether checkout is available when the cart is valid.	1. Add an available product. 2. Open the cart. 3. Inspect the checkout button.	The checkout action is visible and enabled when the cart satisfies the requirements for proceeding.	P0
TC-CART-043	The cart contains at least one valid, purchasable product.	Verify navigation to checkout page	Check whether the checkout button opens the correct destination.	1. Open the cart. 2. Click Checkout.	The user is directed to the intended checkout page or checkout flow. The destination loads correctly without an unexpected error.	P0
TC-CART-044	The cart is empty.	Verify checkout behavior for an empty cart	Ensure users cannot proceed with an invalid empty cart.	1. Remove all products. 2. Open the cart. 3. Inspect and attempt to use the checkout action.	Checkout is disabled, hidden, or handled by an explicit empty-cart redirect, according to the application's requirements. An empty order cannot be submitted.	P0

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-045	The cart contains Product A with quantity 2 and Product B with quantity 1.	Verify cart contents in checkout	Ensure product details are carried forward correctly.	1. Add Product A and Product B. 2. Set the quantities as specified. 3. Open the cart. 4. Proceed to checkout.	The checkout flow reflects the correct products, variants, quantities, and applicable pricing. No item is unexpectedly missing or duplicated.	P0
TC-CART-046	The user is not authenticated, and checkout requires login. Products are in the cart.	Verify login requirement during checkout	Check the authentication flow for guest users.	1. Add products as a guest. 2. Click Checkout. 3. Complete the required login.	The user is directed through the required authentication flow. After login, the application returns the user to the appropriate checkout step and preserves the cart according to its rules.	P1
TC-CART-047	The cart contains products and the checkout page can be opened.	Verify returning from checkout to cart	Check whether users can navigate back and review or modify their cart.	1. Open the cart. 2. Proceed to checkout. 3. Navigate back to the cart using	The cart displays the correct contents and totals. Previously entered	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
				the supported navigation.	information is handled according to the application's session and checkout rules.	
TC-CART-048	The cart is valid and contains purchasable products.	Verify repeated checkout clicks	Ensure rapid repeated clicks do not cause inconsistent navigation or actions.	1. Open the cart. 2. Click Checkout repeatedly in quick succession. 3. Observe navigation and application requests.	The application handles repeated clicks safely. It does not create duplicate transactions or duplicate orders merely because the checkout action was triggered repeatedly.	P1
TC-CART-049	Products are in the cart, and a checkout service failure can be simulated.	Verify checkout error handling	Check behavior when the checkout page or service is temporarily unavailable.	1. Add products to the cart. 2. Simulate a checkout navigation or service error. 3. Click Checkout.	An appropriate error message or recovery option is shown. The cart is not accidentally cleared, and its contents remain consistent when the user retries.	P1

9. Cart Persistence and Session Management

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-050	Product A is in the cart with quantity 2.	Verify cart persistence after page refresh	Check whether cart data survives refreshing the page.	1. Add Product A and set quantity to 2. 2. Refresh the browser. 3. Reopen the cart.	The cart follows the defined persistence policy. For a persistent cart, Product A and its quantity remain correct after refresh.	P1
TC-CART-051	Products are in the cart.	Verify cart persistence during site navigation	Ensure navigating within the site does not unexpectedly clear the cart.	1. Add products. 2. Navigate to a different page on the site. 3. Return to the cart.	The cart contents remain consistent according to the application's persistence rules. Unintended item loss or duplication does not occur.	P1
TC-CART-052	A guest user has products in the cart. The application	Verify guest cart persistence within a session	Ensure a guest can continue using the cart during a	1. Add products as a guest. 2. Navigate to other pages. 3.	The guest cart is preserved for the duration and scope specified by	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
	supports guest carts.		browsing session.	Return to the cart.	the application's guest-cart policy.	
TC-CART-053	A guest cart contains products, and the user can sign in.	Verify guest cart merge after login	Validate the handling of an existing guest cart during authentication.	1. Add Product A as a guest. 2. Sign in to an account that already has Product B in its saved cart. 3. Open the cart.	The guest and account carts are combined or reconciled according to documented merge rules. Products are not unintentionally lost or duplicated, and quantity and stock limits are respected.	P1
TC-CART-054	An authenticated user has products in the cart.	Verify cart behavior after logout	Ensure cart data is handled correctly when the user signs out.	1. Sign in. 2. Add products. 3. Log out. 4. Open the cart or sign in again as appropriate.	Cart data is handled according to the application's authentication and persistence policy. Private account data is not exposed	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
					to another user.	
TC-CART-055	Cart persistence has a configured expiration policy.	Verify cart expiration behavior	Test the handling of carts after their expiration period.	1. Add products. 2. Simulate the configured expiration period or expired-cart state. 3. Reopen the cart.	The expired cart follows the defined policy. The application does not present stale data as guaranteed purchasable, and any cleared or retained data is handled consistently.	P2
TC-CART-056	The same cart is open in two browser tabs, and the application supports multi-tab synchronization.	Verify cart synchronization across tabs	Check for consistent cart state when the cart changes in more than one tab.	1. Open the cart in two tabs. 2. Remove a product in the first tab. 3. Observe the second tab. 4. Refresh or synchronize as required.	The application follows its synchronization policy. When synchronization is supported, changes are reflected without incorrect totals, unintended duplication, or stale state	P2

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
					being treated as current.	

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10. Error Handling, Accessibility, and Responsive Behavior

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-057	The cart contains a product. Network interruption can be simulated.	Verify behavior when cart data cannot be loaded	Check cart loading and error handling during a network failure.	1. Simulate a network interruption. 2. Navigate to the cart or refresh it. 3. Observe the page.	The application presents a useful error or retry state. It does not silently display misleading cart contents or incorrectly claim that data was saved.	P1
TC-CART-058	A keyboard is available, and the cart page can be opened.	Verify keyboard navigation in the cart	Test the main cart actions without a mouse.	1. Open the cart. 2. Navigate between controls using the keyboard. 3. Change quantity. 4. Activate Remove and Checkout	All essential controls are keyboard-accessible, focus order is logical, and visible focus indicators are provided. Actions work without	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
				where applicable.	requiring a mouse.	
TC-CART-059	The cart page supports assistive technology.	Verify accessibility labels for cart controls	Ensure quantity and action controls have understandable accessible names.	1. Inspect the quantity input, plus and minus controls, remove buttons, and checkout action with accessibility tools or a screen reader. 2. Navigate through them.	Controls have appropriate accessible names and roles. Product-specific actions can be distinguished, and relevant state changes are conveyed accessibly.	P1
TC-CART-060	The website supports desktop, tablet, and mobile layouts.	Verify responsive cart layout	Check whether cart content and controls work on different screen sizes.	1. Open the cart on desktop. 2. Repeat on tablet and mobile viewports. 3. Check product details, quantity controls, totals, and checkout.	Cart elements remain usable and readable at supported screen sizes. Important controls are not obscured, and horizontal scrolling or layout issues do not prevent completing cart actions.	P2

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-061	A cart can be populated with many distinct products.	Verify cart behavior with many line items	Test a cart near the application's supported line-item limit.	1. Add many products up to the supported limit. 2. Open the cart. 3. Scroll through the items. 4. Update and remove selected products.	The cart renders correctly within the defined performance requirements. Item information, quantity controls, line-item counts, and totals remain accurate.	P2

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11. Security and Data Integrity

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-062	Products are in the cart. Browser developer tools or request interception are available in the test environment.	Verify server-side price validation	Ensure client-side price changes cannot manipulate the purchase price.	1. Add a product to the cart. 2. Modify any client-side price value or request parameter in the test environment. 3. Attempt to update the cart	The server validates authoritative prices. Modified client-side values are not accepted as the final purchase price, and any discrepancy is handled according to the	P0

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
				or proceed to checkout.	application's rules.	
TC-CART-063	Products are in the cart and request parameters can be inspected in the test environment.	Verify server-side quantity validation	Ensure quantity restrictions cannot be bypassed by manipulating requests.	1. Add a product to the cart. 2. Submit a modified request containing a negative, zero, excessive, or non-integer quantity where the product requires whole units. 3. Check the server response and cart state.	The server rejects invalid quantities, enforces stock and configured quantity limits, and preserves a consistent cart state.	P0
TC-CART-064	Two test accounts are available, and account-specific carts are supported.	Verify cart isolation between users	Ensure one user cannot access or modify another user's private cart.	1. Sign in as User A and add products. 2. Sign out. 3. Sign in as User B. 4. Open the cart and attempt to access any cart resource belonging to User A through the authorized test interface.	User B can access only authorized cart data. User A's private cart cannot be viewed or modified without appropriate authorization.	P0

12. Additional Edge Cases

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
TC-CART-065	A product is in the cart. The product can be deactivated in the catalog.	Verify behavior when a product is discontinued	Check whether a product that is removed from the catalog can still be purchased from the cart.	1. Add Product A to the cart. 2. Deactivate or discontinue it in the test environment. 3. Refresh the cart. 4. Attempt checkout.	The cart follows the discontinued-product policy. The product is not silently treated as purchasable if the application no longer allows its sale. An actionable message or supported recovery option is provided.	P1
TC-CART-066	The cart contains a product with a selected variant.	Verify retaining selected variant after quantity update	Ensure a quantity change does not alter the selected product variant.	1. Add Product A in Red. 2. Set its quantity to 2. 3. Refresh the cart. 4. Inspect the selected variant.	The correct variant remains associated with the line item, and its quantity and pricing are preserved according to the cart persistence rules.	P1
TC-CART-067	A product has a configured purchase	Verify per-customer purchase limits	Check limits that depend on previous purchases or	1. Sign in as a customer subject to the purchase limit. 2. Add the	The application enforces the configured purchase limit and clearly	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
	limit per customer.		customer eligibility.	eligible product. 3. Set the quantity above the permitted limit. 4. Attempt to update the cart or proceed to checkout.	explains any restriction. A client-side change cannot bypass the server-side rule.	
TC-CART-068	The application supports product bundles or quantity-based discounts.	Verify discount recalculation after quantity changes	Check whether threshold-based discounts are recalculated when quantity changes.	1. Add a product eligible for a quantity-based discount. 2. Set the quantity below the discount threshold. 3. Increase the quantity above the threshold. 4. Reduce it below the threshold again.	The applicable discount is correctly applied or removed at each threshold. The displayed line subtotal, cart subtotal, and total remain consistent.	P1
TC-CART-069	Product A and Product B are in the cart, and the application supports a remove-all	Verify clearing the entire cart	Ensure a supported clear-cart action removes all line items.	1. Add multiple products. 2. Activate the clear-cart action. 3. Confirm the action if prompted. 4.	All intended line items are removed. The empty-cart state is shown, and the line-item count, quantity	P1

Test Case ID	Pre-condition	Test Case Name	Description	Steps	Expected Result	Priority
	action if specified.			Reopen or refresh the cart.	indicator, and totals are updated correctly.	
TC-CART-070	The application supports cart recovery after a temporary server or connectivity issue.	Verify recovery after retrying a failed cart update	Ensure a failed update can be retried without duplicating changes.	1. Add a product to the cart. 2. Simulate a temporary failure during a quantity update. 3. Restore connectivity. 4. Retry the update. 5. Refresh the cart.	The final quantity and totals match the successfully saved state. The retry does not create duplicate updates or leave conflicting client-side and server-side values.	P1

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13. Recommended Test Execution Order

For a practical test cycle, execute the cases in the following order:

Execution Phase	Test Case Range	Purpose
Phase 1: Smoke Testing	TC-CART-001, TC-CART-009, TC-CART-017, TC-CART-029, TC-CART-042, TC-CART-043	Confirm that core cart operations and checkout navigation work.
Phase 2: Functional Testing	TC-CART-002 to TC-CART-028	Verify line items, quantity changes, product information, prices, and calculations.

Execution Phase	Test Case Range	Purpose
Phase 3: Removal and Inventory	TC-CART-029 to TC-CART-041	Validate product deletion, stock handling, and availability rules.
Phase 4: Checkout and Persistence	TC-CART-042 to TC-CART-056	Verify checkout navigation, login flows, and cart persistence.
Phase 5: Nonfunctional Testing	TC-CART-057 to TC-CART-061	Evaluate error handling, accessibility, responsiveness, and performance.
Phase 6: Security and Edge Cases	TC-CART-062 to TC-CART-070	Validate server-side enforcement, data integrity, and less common scenarios.

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14. Entry and Exit Criteria

Entry criteria

Testing can begin when the application is accessible, core cart functionality is available, product data has been prepared, and the test environment can simulate relevant inventory and server conditions.

Exit criteria

Cart testing can be considered complete when all planned test cases have been executed, all critical and high-priority failures have been resolved or formally accepted, expected results have been verified, and the remaining defects have been documented.

Actual pass/fail status should be recorded only after executing each test case against the application.

Total coverage: 70 test cases.

The suite covers the primary shopping cart requirements and extends into inventory changes, pricing integrity, session behavior, accessibility, and security. Before execution, align policy-dependent expectations—such as maximum quantities, cart persistence, variant handling, and empty-cart checkout behavior—with the application's actual requirements.